



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8393**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	3DS	Seat No:	2022700060
Course:	Section:1				

L_41_673_986_L_1505_41477



Name of the candidate. siddhark thikekar

Candidate's UID number:

2	0	2	2	7	0	0	0	6	0
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Examination class room number:

3	0	5
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Invigilator's signature with date:

	8/10
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(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

PROGRAM: CSE DS

DAY AND DATE: Wednesday 8/10/25

COURSE NAME: Deep learning

SEMESTER :

I / II / III / IV / V / VI / VII / VIII ✓

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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(Begin answer for each question on a new page)

Q.26) Proof :- Autoencoder trained with MSE is equivalent to PCA

$d(n) < d(n)$ $\hat{n}$ (output layer)

$\Rightarrow H = W X$

$\Rightarrow \hat{X} = W^* H$

n (input)

It is equal to PCA if :-

- 1) Linear Encoder
- 2) Linear decoder
- 3) Squared error loss function
- 4) Inputs are normalized

$$\Rightarrow n_{ij}'' = \frac{1}{\sqrt{m}} \left(n_{ij} - \frac{1}{m} \sum_k (n_{jk}) \right)$$

mean

X'

Scaling with mean

$$X = \frac{1}{\sqrt{m}} X'$$

From PCA $V = X^T X$ - ①

substituting :-

$$\left(\frac{1}{\sqrt{m}} X' \right)^T \left(\frac{1}{\sqrt{m}} X \right)$$

$$\Rightarrow \frac{1}{m} (X')^T (X)$$

Autoencoder minimizes $\Rightarrow \min (n_{ij} - \hat{x})^2$

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Equations :-

 $X \rightarrow$ input $H \rightarrow$ hidden layer $W^* \rightarrow$ hidden layer weights $\hat{X} \rightarrow$ Output $W \rightarrow$ input layer weights

$$\hat{X} = W^* H$$

$$H = X W$$

From SVD :- $H = U \Sigma V^T$

$$\text{Equating :- } X W = U \Sigma V^T$$

$$X = U \Sigma$$

 $\downarrow$
Output
of Encoder

$$W = V^T$$

 $\downarrow$
Output of
decoderMultiplying $(X X^T) (X X^T)^{-1}$ into $X = U \Sigma$

$$H = (X X^T) (X X^T)^{-1} U \Sigma$$

$$\boxed{\{X X^T\} = I}$$

$$\boxed{X X^{-1} = I}$$

Substituting $X = U \Sigma V^T$

$$H = (X (U \Sigma V^T)^T) ((U \Sigma V^T) (U \Sigma V^T)^T)^{-1} U \Sigma$$

$$H = (X V \Sigma^T U^T) \left(\underbrace{U \Sigma V^T V \Sigma^T U^T}_I \right)^{-1} U \Sigma$$

$$H = (X V \Sigma^T U^T) \left(U \Sigma \Sigma^T U^T \right)^{-1} U \Sigma \quad \left\{ \begin{array}{l} U^T U = I \\ V^T V = I \end{array} \right.$$

$$H = \underbrace{(X V \Sigma^T)}_I U^T U \left(\Sigma \Sigma^T \right)^{-1} \underbrace{U^T U}_I \Sigma$$

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As $(UU^T)^{-1} = U^T U$ as U is orthogonal.

$$H = (XV \Sigma^T) (\Sigma \Sigma^T)^{-1} \Sigma$$

$$H = X \underbrace{V \Sigma^T \Sigma}_I \underbrace{\Sigma^{-1} \Sigma}_I$$

$$H = XV$$

∴ From this we get linear dependency of H to X and

$$\boxed{V=W}$$

From SVD $V = X^T X$

From PCA covariance matrix = $X^T X$

∴ This shows that encoder and Projection of PCA are Equal

⇒ Overcomplete autoencoder tends to learn the identity function as the hidden layer has more no. of neurons ~~the~~ ~~the~~ ~~the~~ than the input. So, because of this the input is straight away copied from in → hidden → output without the algorithm learning anything.

So, we need to apply regularization techniques to this autoencoder such as denoising the input, so that the model learns from input rather than memorizing it.

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Q.1a) $f(w) = \frac{1}{2} \|xw - y\|^2$

Proof (Derivation of gradient update rule)

$\Delta\theta = \eta u$ (Taking η small steps in u direction)

~~$L(\theta)$~~ Taylor Series Expansion:-

$$L(\theta + \eta u) = L(\theta) + \eta u^T \nabla_{\theta} L(\theta) + \frac{\eta^2}{2!} \dots$$

We want :- Small

$\Rightarrow L(\theta + \eta u) - L(\theta) < 0$ (To learn from input)

$\Rightarrow \boxed{u^T \nabla_{\theta} L(\theta) < 0}$

Taking an angle between u & $\nabla_{\theta} L(\theta)$

$$1 \leq \cos \beta = \frac{u^T \nabla_{\theta} L(\theta)}{\|u\| \|\nabla_{\theta} L(\theta)\|} \leq -1$$

Multiplying by K :-

$$K \leq K \cos \beta = u^T \nabla_{\theta} L(\theta) \leq -K$$

$\Rightarrow K \cos \beta = u^T \nabla_{\theta} L(\theta) = L(\theta + \eta u) - L(\theta)$

$\hookrightarrow$ we get smallest value at

$\beta = 180^\circ$, $\cos \beta = -1$

$\therefore$ ~~gradient~~ ~~learn~~ learns by going opposite to gradient



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$$w_{t+1} = w_t - \eta \nabla w$$

$$b_{t+1} = b_t - \eta \nabla b$$

$$\nabla w = \frac{\partial L(w, b)}{\partial w} \quad \nabla b = \frac{\partial L(w, b)}{\partial b}$$

Error function :- $\frac{1}{2} (f(w) - y)^2$

Here $f(w) = f(w) = \frac{1}{2} \|Xw - y\|^2$

Taking Partial derivative :-

$$\nabla_w = \frac{1}{2} \times 2 (f(w) - y) \underbrace{\frac{\partial}{\partial w} (f(w) - y)}_{(2)}$$

$$\nabla_w = (f(w) - y)$$

Solving Equ (2) :-

$$\frac{\partial}{\partial w} (f(w) - y) \Rightarrow \frac{\partial}{\partial w} (f(w))$$

Substituting $f(w)$:-

$$\frac{\partial}{\partial w} \left(\frac{1}{2} \|Xw - y\|^2 \right) \Rightarrow$$

$$\frac{1}{2} \times 2 \|Xw - y\| \times (X - y)$$

$$\Rightarrow (Xw - y) (X - y)$$

Putting in original Equ :-

$$\nabla_w = (f(w) - y) (Xw - y) (X - y)$$



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Q.16) model capacity too high means that the model is very capable to understand complex input, but can overfit ~~but~~ due to memorization rather than learning from it. Due to this the model tends to have poor generalisation as when the model overfits it performs very well on training data but fails to give results on test data due to memorization of the inputs. solution can be using regularization techniques ~~so~~ which forces model to learn rather than memorizing inputs by adding a penalty terms. these terms help model to learn as they push the model or penalize it when it ~~tends~~ to ~~learn~~ from the input tends to memorize rather than learning. This helps model to perform way better and helps in generalisation of it. Too high capacity models also have many complex layers so minimizing them can also solve the generalization problem.



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Q2a) input size :- $H \times W \times C$

filters is K :- $F \times F$

$$\text{Output size :- } \left(\frac{H - F + 2P + 1}{S} \right) \times \left(\frac{W - F + 2P + 1}{S} \right) \times K$$

depthwise separable convolutions are critical in modern architecture as it helps in finding features from multiple layer. Each layer extracts some important feature from the input such as edges, texture, shapes etc. At the end when input passes through these layers each of these results are combined and an output is produced according to it.

output size :-

$$\left(\frac{H - F + 2P + 1}{S} \right) \times \left(\frac{W - F + 2P + 1}{S} \right) \times K$$

where P is the padding and S is stride of the filter.

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Q3a) Normally RNN's have a very short memory and are used for sequential data inputs. It has a very short memory means it can't remember very earlier content from a sentence for example. It also suffers from short memory due to vanishing gradient problem where in the weights if repeatedly multiplied by number less than 1 tend to stop the updation of the weight totally and model cannot learn further as $W \approx 0$ (learning). LSTM solves this problem as it has gates :-

- 1) forget gate
- 2) input gate
- 3) Output gate

It can decide how much content to ~~remember~~ remember from the past through help of gates. forget gate helps in forgetting or removing previous content, input takes or stores new things in memory and output gate gives output by seeing previous hidden layer output and input.

$$\text{input} :- \sigma_i(W_i[N_{i-1}, n], b_i)$$

$$\text{output} :- \sigma_o(W_o[N_{i-1}, n], b_o)$$

$$\text{forget gate} :- \sigma_f(W_f[N_{i-1}, n], b_f)$$

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Q36) GRU Equations :-

Update gate :- $\sigma_u(W_u[n_{i-1}, x])$

Reset gate :- $\sigma_r(W_r[n_{i-1}, x])$

Candidate gate :- $\tanh(W_c[n_{i-1}, x])$

Grated Recurrent Units (GRU) tend to have fewer gates than LSTM but perform Equally well for an input. It removes and takes input in one gate only (Update gate) which lowers the number of gate as compared to LSTM. Candidate gate is the output of the previous hidden layer which helps in remembering content from the past and give output according to it.

Reset gate helps in removing things from the past contents. In real world it would take much less space than LSTM.

[illegible]

[illegible]

[illegible]

[illegible]